US MONETARY POLICY, MARKET-IMPLIED SHORT-RATE CONDITIONS, AND THE 1-MONTH TREASURY YIELD

A Nelson-Siegel-Svensson and Stability-Aware Ridge Forecasting Study

PROJECT OBJECTIVE

- Measure the gap between the Treasury curve's fitted instantaneous short rate and the Federal funds target midpoint.
- Test whether policy-spread and short-rate information improves next-month 1-month Treasury-yield forecasts relative to a no-change benchmark.

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30 October 2026

Data: Federal Reserve Economic Data (FRED). Treasury and target-rate series: Board of Governors of the Federal Reserve System.

IMPORTANT INTERPRETIVE NOTE

- The policy spread measures a fitted Treasury short-rate gap, not a pure market expectation of the next Federal Reserve decision.
- The reported out-of-sample advantage is modest and should be interpreted as sample-specific incremental accuracy, not definitive evidence of superior forecasting power.

Executive summary

This study links the US Treasury yield curve with Federal Reserve policy and evaluates whether derived short-rate information improves forecasts of the next-month 1-month Treasury yield. Monthly Nelson-Siegel-Svensson (NSS) curves provide a fitted instantaneous short rate. The difference between this fitted rate and the Federal funds target midpoint is defined as the policy spread.

$$Policy\ spread_t = NSS\ fitted\ short\ rate_t - Federal\ funds\ target\ midpoint_t$$

Part II forecasts the next-month change in the observed 1-month Treasury yield using a rolling Ridge regression. The final specification combines yield momentum, the change in the policy midpoint, the policy spread, and the change in the policy spread. The model is evaluated against a no-change benchmark, under which the current yield is used as the next-month forecast.

Metric	Ridge model	No-change benchmark	Improvement
RMSE	0.201 pp (20.1 bps)	0.207 pp (20.7 bps)	2.589%
MAE	0.111 pp (11.1 bps)	0.112 pp (11.2 bps)	0.844%

MAIN CONCLUSION

- The Ridge model produced slightly lower out-of-sample RMSE and MAE than the no-change benchmark.
- The improvement is economically modest, confirming that the current 1-month yield remains a difficult benchmark to beat.
- The results are consistent with policy and NSS-derived indicators containing incremental information in this sample, but statistical significance is not established in the current study.

1. Research design

1.1 Part I: fitted short-rate conditions

Part I compares the instantaneous short rate implied by the fitted Treasury curve with the Federal funds target midpoint. The objective is to assess whether fitted Treasury short-rate conditions are above, below, or broadly aligned with the contemporaneous policy setting.

1.2 Part II: next-month yield forecast

Part II forecasts the next-month change in the observed 1-month Treasury yield. Candidate predictors are evaluated using time-ordered validation, and the locked final specification is compared with a no-change benchmark in the out-of-sample period.

2. Data

Daily Treasury constant-maturity yields and Federal funds target-range data are retrieved through the FRED API and converted to month-end observations using the final available value in each month. The target midpoint is calculated as the average of the upper and lower limits of the target range.

$$Policy\ midpoint_t = \frac{(Upper\ limit_t + Lower\ limit_t)}{2}$$

Data block	Content
Treasury curve	1M, 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, and 30Y constant-maturity yields
Policy	Federal funds target upper limit, lower limit, and calculated midpoint
Frequency	Daily observations converted to month-end
Study period	January 2009 to October 2026

DATA CAUTION

- Month-end Treasury yields and the target midpoint are contemporaneous observations. The design does not claim that the fitted spread isolates expectations from liquidity, supply, convenience-yield, or curve-fitting effects.

3. NSS curve and policy spread

3.1 NSS specification

$$y_t(\tau) = \beta_{0,t} + \beta_{1,t}\left(\frac{1-e^{-\tau/\lambda_{1,t}}}{\tau/\lambda_{1,t}}\right) + \beta_{2,t}\left(\frac{1-e^{-\tau/\lambda_{1,t}}}{\tau/\lambda_{1,t}} - e^{-\tau/\lambda_{1,t}}\right) + \beta_{3,t}\left(\frac{1-e^{-\tau/\lambda_{2,t}}}{\tau/\lambda_{2,t}} - e^{-\tau/\lambda_{2,t}}\right)$$

The NSS model summarizes the yield curve through a long-run level, a slope component, and two curvature components. The fitted parameters vary by month.

3.2 Implied instantaneous short rate

$$r_t(0) = \lim_{\tau \to 0} y_t(\tau) = \beta_{0,t} + \beta_{1,t}$$

This quantity is a theoretical instantaneous Treasury rate derived from the fitted curve. It is not an observed overnight policy rate and should not be interpreted as a direct measure of a specific future Federal Reserve action.

3.3 Policy spread and interpretation

$$s_t = r_t(0) - \mathit{Policy\ midpoint}_t$$

Spread	Careful interpretation
Positive	The fitted instantaneous Treasury short rate is above the contemporaneous policy midpoint. This is consistent with tighter market-implied short-rate conditions, but may also reflect liquidity, supply, convenience-yield, or fitting effects.
Near zero	The fitted Treasury short rate is broadly aligned with the current policy midpoint.
Negative	The fitted instantaneous Treasury short rate is below the policy midpoint. This is consistent with easier market-implied short-rate conditions, subject to the same caveats.

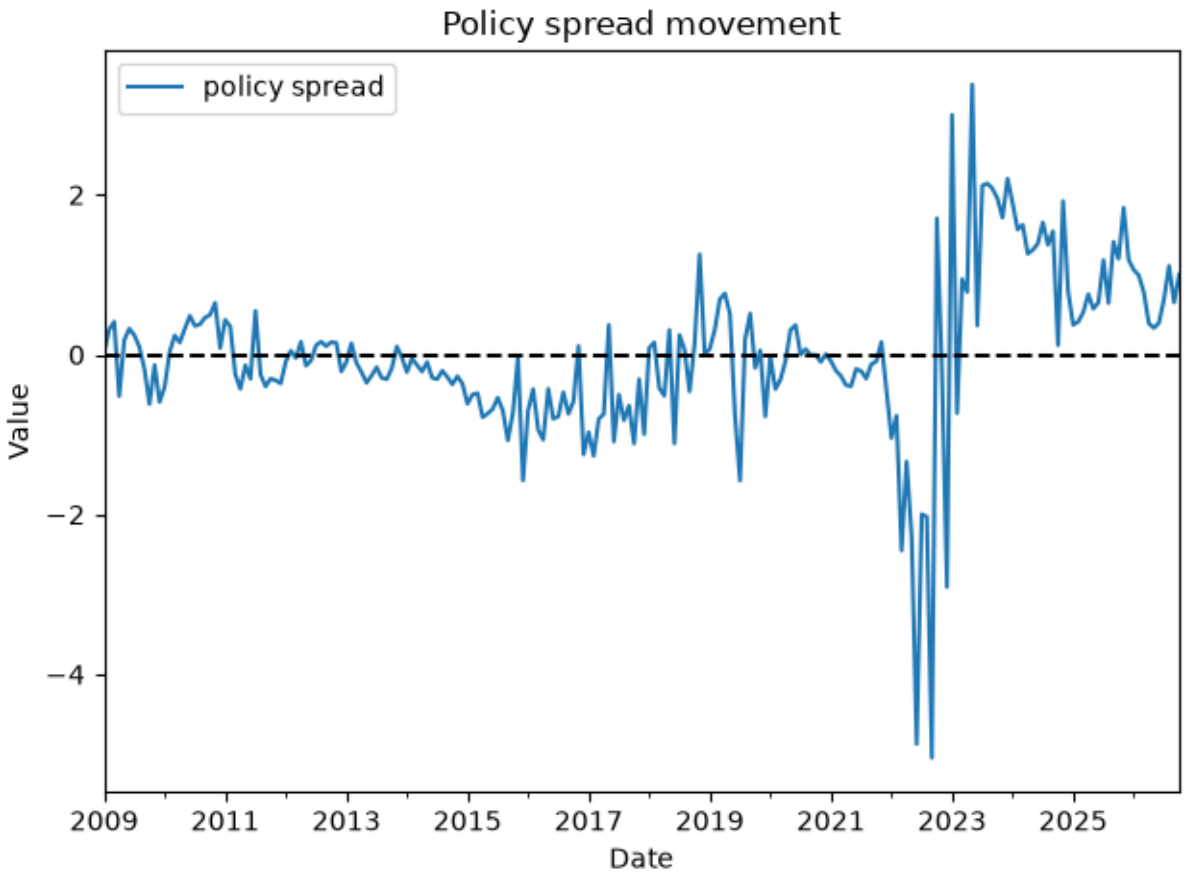


Figure 1. Policy-spread movement over time. Source: Author calculations using FRED data.

The policy spread became visibly more volatile after 2021 and was mostly positive from 2023 onward. The subsequent narrowing is consistent with partial realignment between the fitted Treasury short rate and the policy midpoint. However, visual convergence toward zero is not sufficient evidence of statistical mean reversion. Formal unit-root and half-life tests are required before making that claim.

Macroeconomic developments such as growth surprises, inflation surprises, FOMC communication, Treasury bill supply, and geopolitical events are possible hypotheses for spread movements. They are not tested as causal drivers in this study.

4. Forecasting methodology

4.1 Target variable and benchmark

$$\Delta y_{t+1}^{1M} = y_{t+1}^{1M} - y_t^{1M}$$

where:

- Δy_{t+1}^{1M}*: change in the 1-month Treasury yield from month* t *to month* $t+1$
- y_t^{1M}*: 1-month Treasury yield in the current month*
- y_{t+1}^{1M}*: 1-month Treasury yield in the next month*

The model predicts the next-month change in the observed 1-month Treasury yield. The final level forecast adds the predicted change to the current observed yield. The no-change benchmark is:

$$\hat{y}_{t+1}^{benchmark} = y_t^{1M}$$

4.2 Candidate predictor groups

Group	Economic content
Momentum	Current and lagged observed 1-month yield changes
Smoothed momentum	Current change, 3-month mean change, and recent volatility
Policy and momentum	Yield momentum, policy-midpoint change, and policy spread
Policy dynamics	Policy-midpoint change, policy spread, and policy-spread change
NSS short-rate dynamics	Yield momentum, NSS short-rate change, and policy spread

4.3 Locked final specification

FINAL OUT-OF-SAMPLE MODEL

- Predictors: current 1-month yield change, policy-midpoint change, policy spread, and policy-spread change.
- Training window: rolling 48 months.
- Ridge penalty alpha: 180.
- Forecast shrinkage weight: 0.25.
- Signal threshold: 1.675 basis points.
- Standardization: fitted using historical training observations only at each forecast date.

This table defines the single model used for the reported out-of-sample results. Earlier validation outputs from alternative predictor groups, window structures, weights, or thresholds are intentionally excluded to prevent specifications from being mixed.

4.4 Rolling Ridge estimation

At each forecast date, predictors are standardized using only the corresponding historical training window. Ridge regression minimizes squared forecast error plus an L2 penalty on the slope coefficients:

$$\min_{\beta_0,\beta} \sum_{i=1}^{n} \left[\Delta y_{i+1}^{1M} - \left(\beta_0 + \sum_{j=1}^{p} \beta_j Z_{i,j} \right) \right]^2 + 180 \sum_{j=1}^{p} \beta_j^2$$

where:

- Δy_{i+1}^{1M} *is the actual next-month change in the 1-month Treasury yield.*
- β_0 *is the intercept.*
- β_j *is the coefficient for predictor* j*.*
- $Z_{i,j}$ *is the standardized value of predictor* j *for observation* i*.*
- p *is the number of predictors.*
- n *is the number of observations in the historical training window.*
- $\alpha = 180$ *controls the strength of the L2 penalty.*

The penalty reduces coefficient instability when the predictors are correlated or the rolling sample is limited. The intercept is not penalized.

4.5 Forecast shrinkage and signal threshold

Write the formula as:

$$\widehat{\Delta y}_{t+1}^{weighted} = 0.25 \times \widehat{\Delta y}_{t+1}^{raw}$$

where:

- $\widehat{\Delta y}_{t+1}^{raw}$ *is the original predicted change in the 1-month Treasury yield.*
- $\widehat{\Delta y}_{t+1}^{weighted}$ *is the adjusted prediction.*
- 0.25 *is the forecast weight.*

The raw forecast is shrunk toward zero. The weighted forecast is retained only when its absolute magnitude is at least 1.675 basis points; otherwise, the model reverts to a no-change forecast.

$$\widehat{\Delta y}_{t+1}^{used} = \begin{cases} \widehat{\Delta y}_{t+1}^{weighted}, & \text{if } \left|\widehat{\Delta y}_{t+1}^{weighted}\right| \geq 0.01675 \text{ percentage points} \\ 0, & \text{otherwise} \end{cases}$$

$$\hat{y}_{t+1}^{1M} = y_t^{1M} + \widehat{\Delta y}_{t+1}^{used}$$

MODEL-SELECTION LIMITATION

- The predictor set, alpha, shrinkage weight, threshold, and window length must be selected without using the final test outcomes. Because the current report does not document a nested validation audit or the number of specifications evaluated, residual data-mining risk remains.

5. Out-of-sample results

Metric	Ridge model	No-change benchmark	Improvement
RMSE	0.201 pp (20.1 bps)	0.207 pp (20.7 bps)	2.589%
MAE	0.111 pp (11.1 bps)	0.112 pp (11.2 bps)	0.844%

The Ridge model generated slightly lower point estimates of RMSE and MAE than the no-change benchmark in the reported test sample. The advantage is small, so the result should be described as modest incremental accuracy rather than conclusive forecasting superiority.

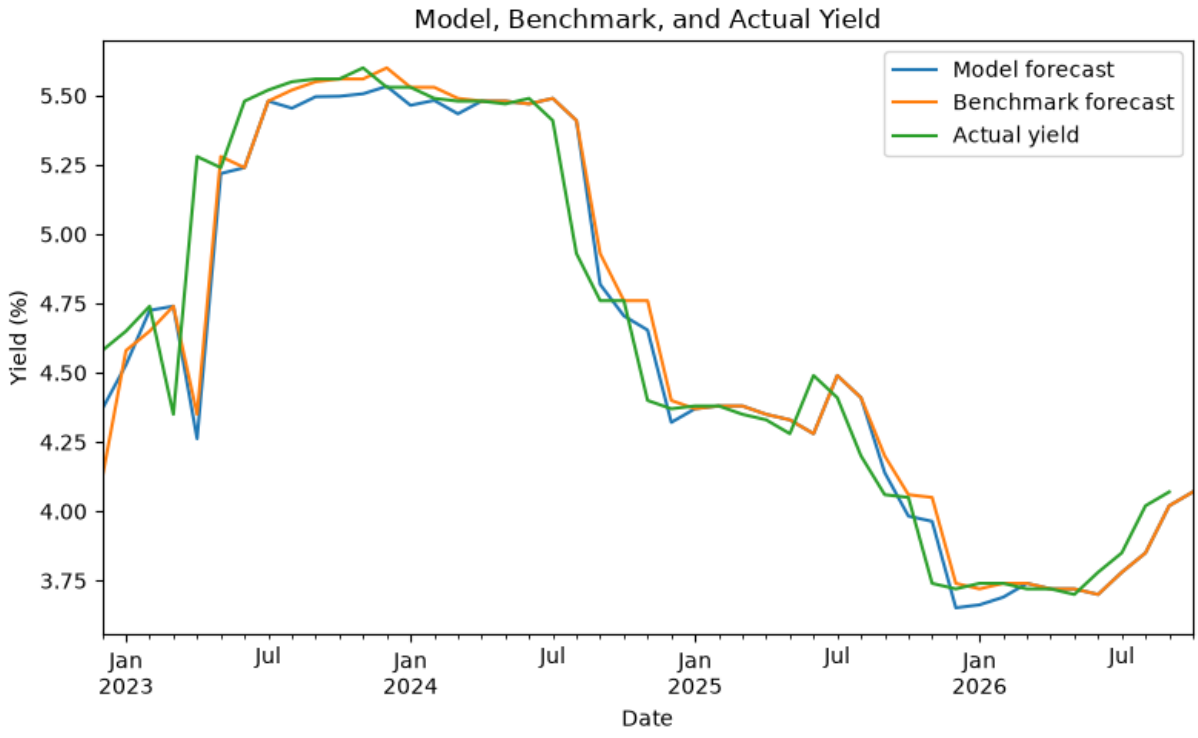


Figure 2. Actual 1-month yield, Ridge forecast, and no-change benchmark. Source: Author calculations using FRED data.

The model and benchmark track the actual yield closely because the 1-month yield is highly persistent. The level chart therefore provides limited visual evidence about the source of the small error improvement. A cumulative squared-error difference or rolling RMSE chart would provide a more diagnostic comparison.

6. Forecast-error analysis

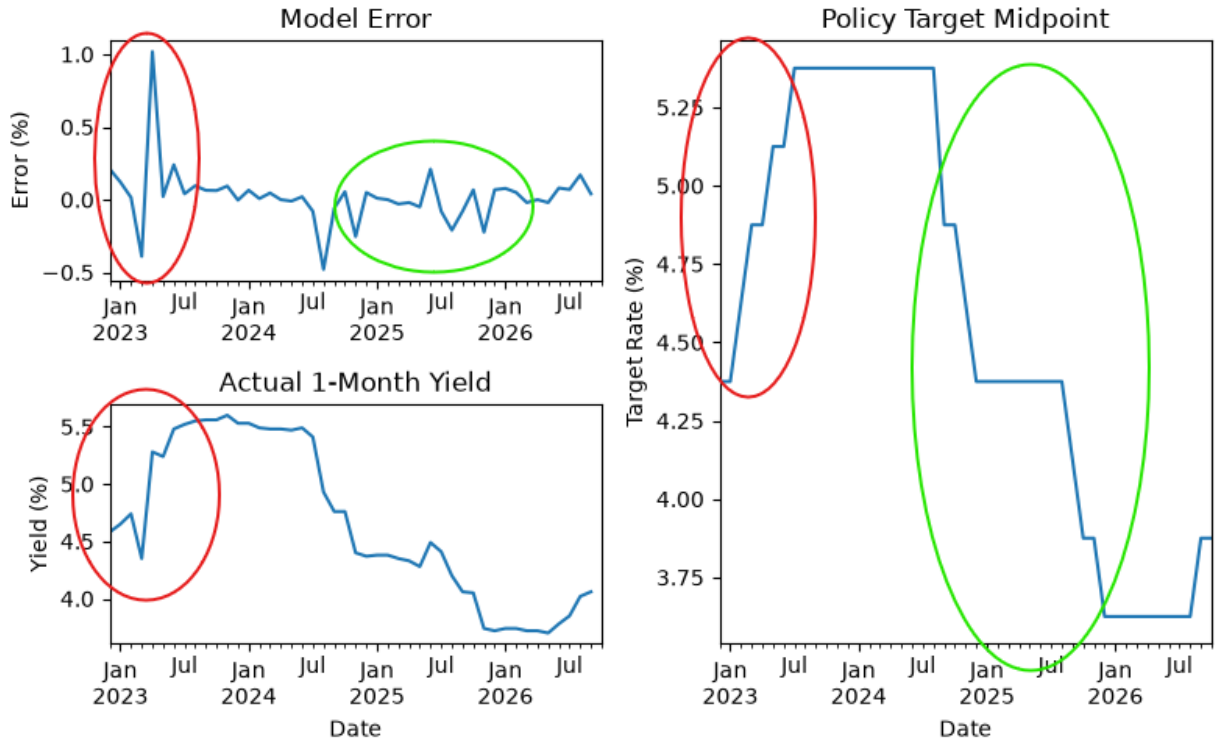


Figure 3. Model error, policy target midpoint, and actual 1-month yield. Source: Author calculations using FRED data.

The largest absolute forecast error occurred in early 2023, when the 1-month Treasury yield adjusted abruptly during a period of rapid policy tightening. The maximum error was approximately 1 percentage point, or 100 basis points. Errors were generally smaller during the smoother decline in the policy midpoint and 1-month yield from late 2024 to early 2026.

The pattern is consistent with, but does not prove, two hypotheses: the model may perform better during gradual easing than during abrupt tightening, or the model may capture persistent policy-driven trends more effectively than turning points. Regime-specific performance should be calculated before accepting either explanation.

Recommended diagnostic	Purpose
Tightening vs easing RMSE/MAE	Test whether forecast accuracy differs by policy regime
High- vs low-volatility months	Assess sensitivity to short-rate instability
Cumulative squared-error difference	Identify when the model adds or loses value relative to the benchmark
Directional accuracy	Measure whether the model predicts the sign of monthly changes
Threshold activation rate	Show how often the model differs from the no-change benchmark

7. Robustness, limitations, and next tests

7.1 Statistical inference

Lower point estimates of RMSE and MAE do not establish that forecast performance is statistically different. A Diebold-Mariano test can compare loss differentials, while a Clark-West test may be useful when evaluating a

larger predictive model against a nested benchmark. A block bootstrap can provide confidence intervals that account for serial dependence.

7.2 Policy-spread properties

The apparent return of the spread toward zero should be evaluated using Augmented Dickey-Fuller, Phillips-Perron, and KPSS tests. An AR(1) model can be used to estimate persistence and an implied half-life. Structural-break tests are also important because spread volatility increased sharply after 2021.

7.3 Incremental value of the policy spread

The key research question requires an ablation test. The final model should be compared with an otherwise identical Ridge model that excludes the policy spread and its change. If forecast accuracy does not deteriorate materially, the spread may not provide incremental information beyond yield and policy momentum.

7.4 Scope limitations

- The NSS instantaneous short rate is model-implied and sensitive to curve fit, parameter bounds, and short-end data quality.
- Treasury yields and the Federal funds target midpoint represent different instruments and may embed different liquidity and institutional effects.
- The economic value of a 0.6 basis-point RMSE reduction depends on the intended decision or trading application and associated transaction costs.

8. Conclusion

This project develops a coherent two-stage framework. Monthly NSS curve fitting produces an economically interpretable, model-implied instantaneous Treasury short rate and a policy spread relative to the Federal funds target midpoint. A rolling Ridge model then evaluates whether yield-momentum and policy-related indicators improve next-month forecasts of the observed 1-month Treasury yield.

In the reported out-of-sample period, the final model achieved an RMSE of 0.201 percentage points and an MAE of 0.111 percentage points, compared with 0.207 and 0.112 for the no-change benchmark. The model therefore generated slightly lower forecast errors, but the improvement was modest.

Policy and NSS-derived indicators may contain incremental forecasting information in this sample, while the current yield remains a powerful benchmark.

FINAL TAKEAWAY

- The project is best interpreted as evidence of a small, conditional forecasting advantage, not a definitive rejection of the no-change benchmark.